

Mathematical Foundations & Acoustic Physics Specification

Rigorous Signal Processing Formulations, Biometric Equations, and Derivations

Problem Statement:	AI-Powered Real-Time Voice Cloning Detection & Impersonation Defense (PS-26104)	Document Ref:	VS-DOC-2026-01
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1. Digital Acoustic Discretization & Time-Frequency Representation

VoiceShield AI ingests audio at a standardized sampling frequency $f_s = 16,000$ Hz (Nyquist frequency $f_N = 8,000$ Hz), capturing human speech fundamentals (80–300 Hz) and high-frequency vocoder transition dispersion (3,500–7,500 Hz). The continuous waveform $x(t)$ is digitized to discrete sequence $x[n] = x(n \cdot T_s)$ with 16-bit PCM quantization.

The Short-Time Fourier Transform (STFT) splits the signal into overlapping frames using a 25ms Hamming window ($N = 400$ samples) with a 10ms frame hop ($H = 160$ samples):

$$X(m, \omega) = \sum_{n=-\infty}^{\infty} x[n] \cdot w[n - m] \cdot e^{-j \omega n}$$

Where the symmetric Hamming window minimizes side-lobe leakage across discrete Fourier bins:

$$w[n] = 0.54 - 0.46 \cdot \cos\left(\frac{2\pi n}{N-1}\right), \quad 0 \leq n \leq N-1$$

2. Levinson-Durbin Lattice Solver & Glottal Inverse Residual Extraction

Human speech production is modeled as a source-filter system where the vocal tract acts as an all-pole linear filter $H(z)$ driven by glottal fold excitation $E(z)$. Neural vocoders (Diffusion, HiFi-GAN, WaveGlow) synthesize audio without a biological vocal tract, leaving mathematical artifacts in the inverse residual.

$$H(z) = G / (1 - \sum_{k=1}^p a_k z^{-k})$$

We solve for the 16th-order Linear Prediction Coefficients (LPC, $p=16$) via the Yule-Walker equations using the Levinson-Durbin recursive algorithm in $O(p^2)$ operations instead of $O(p^3)$:

Levinson-Durbin Recursive Algorithm:

- Initialization: $E_0 = R(0)$
- For $i = 1, 2, \dots, p$:
 - $k_i = - [R(i) + \sum_{j=1}^{i-1} a_j^{(i-1)} R(i-j)] / E_{i-1}$
 - $a_i^{(i)} = k_i$
 - $a_j^{(i)} = a_j^{(i-1)} + k_i \cdot a_{i-j}^{(i-1)}$, for $1 \leq j \leq i-1$
 - $E_i = E_{i-1} \cdot (1 - k_i^2)$

The inverse residual signal $e[n]$ represents the glottal volume velocity derivative:

$$e[n] = x[n] - \sum_{k=1}^p a_k \cdot x[n - k]$$

Biometric vocal excitation is verified by measuring the glottal open quotient $OQ = T_{open} / T_0$ and residual impulse kurtosis $\kappa = \mu_4 / \sigma^4$. Synthetic vocoders fail to model biological mucosal wave elasticity, yielding abnormal kurtosis spikes.

3. Modified Group Delay (MGD) Phase Spectrum Formulation

While standard magnitude spectrograms ignore phase information, neural vocoders introduce unnatural phase dispersion during inverse STFT reconstruction. The Group Delay function is the negative derivative of the phase spectrum:

$$\tau(\omega) = - \frac{d \theta(\omega)}{d\omega} = - \frac{d}{d\omega} [\text{Im}(\log X(\omega))]$$

Standard group delay suffers from pitch resonance spikes where $|X(\omega)| \approx 0$. We implement the Modified Group Delay (MGD) with cepstral smoothing parameters $\alpha = 0.4$ and $\gamma = 0.9$:

$$\tau_m(\omega) = [X_R(\omega) Y_R(\omega) + X_I(\omega) Y_I(\omega)] / |S(\omega)|^{2\gamma} \cdot | \tau_m(\omega) / | \tau_m(\omega) | |^{\alpha}$$

Where $Y(\omega) = FFT\{ n \cdot x[n] \}$, and $|S(\omega)|$ is the smoothed spectral envelope obtained via discrete cosine liftering. This isolates high-frequency vocoder phase incoherence.

4. Attentive Statistics Pooling (ASP) Mathematical Derivation

Standard global average pooling discards temporal variation crucial for detecting fleeting voice clone splices. Attentive Statistics Pooling (ASP) computes attention-weighted mean (μ) and attention-weighted standard deviation (σ):

Given frame embeddings $\mathbf{h}_t \in \mathbb{R}^D$ for $t = 1, \dots, T$:
1. Attention score: $e_t = v^T \cdot \tanh(W \cdot \mathbf{h}_t + b)$
2. Normalized weights: $\alpha_t = \exp(e_t) / \sum_{\tau=1}^T \exp(e_\tau)$
3. Weighted Mean: $\mu = \sum_{t=1}^T \alpha_t \cdot \mathbf{h}_t$
4. Weighted Variance: $\sigma = \sqrt{ \sum_{t=1}^T \alpha_t \cdot (\mathbf{h}_t - \mu)^T (\mathbf{h}_t - \mu) }$
5. Output Representation: $\mathbf{e}_{ASP} = [\mu \parallel \sigma] \in \mathbb{R}^{2D}$

Capturing both first-order centroid and second-order dispersion enables the Conformer to capture micro-prosodic tremors that synthetic speech algorithms smooth out.

5. The 7-Vector Acoustic Biometric Formulations

Each acoustic frame is evaluated across 7 physiological dimensions:

Vector Name	Mathematical Formula	Human vs Synthetic Target
1. Spectral Centroid	$C = (\sum k \cdot X[k]) / (\sum X[k])$	Human: 1200–2400 Hz Cloned: Excessive HF (>3200 Hz)
2. Spectral Rolloff	$\sum_{k=0}^{K_r} X[k] = 0.85 \cdot \sum_{k=0}^{N/2} X[k] $	Human: Gradual energy decay Vocoder: Steep cutoff slope
3. Fundamental Pitch (F0)	$F0 = \operatorname{argmax}_{\tau} R_{\{xx\}}(\tau), \tau \in [50, 400] \text{ Hz}$	Human: Dynamic pitch wander Clone: Unnatural monotone lock
4. Micro-Jitter (RAP)	$RAP = [(1/(N-2) \sum T_i - (T_{i-1} + T_{i+1})/3) / T_{mean}] \times 100$	Human: 0.3% – 1.2% Clone: <0.15% (Robotic) or >3.5%
5. Micro-Shimmer (APQ3)	$APQ3 = [(1/(N-2) \sum A_i - (A_{i-1} + A_{i+1})/3) / A_{mean}] \times 100$	Human: 1.5% – 4.0% Clone: Abnormally static amplitude
6. Harmonics-to-Noise (HNR)	$HNR = 10 \cdot \log_{10}[R_{\{xx\}}(T_0) / (R_{\{xx\}}(0) - R_{\{xx\}}(T_0))] \text{ dB}$	Human: 12 – 24 dB Replay/Vocoder: <10 dB (Noise floor)
7. Zero Crossing Rate (ZCR)	$ZCR = 1/(2N) \sum_{n=1}^N \operatorname{sgn}(x[n]) - \operatorname{sgn}(x[n-1]) $	Human: Clean phoneme transitions Synthesized: Glitched unvoiced frames

6. Multi-Model Consensus Dynamic Risk Scoring Matrix

Individual forensic scores are projected through a non-linear sigmoidal risk engine:

$$R_{\text{raw}} = 0.35 \cdot P_{\text{conformer}} + 0.20 \cdot P_{\text{glottal}} + 0.15 \cdot P_{\text{phase}} + 0.15 \cdot P_{\text{voicemod}} + 0.15 \cdot P_{\text{semantic}}$$
$$R_{\text{composite}} = 1.0 / (1.0 + \exp(- 10.0 \cdot (R_{\text{raw}} - 0.50)))$$

- Decision Boundaries Enforced:**
- **Risk < 40%:** *VERIFIED HUMAN (Call allowed, green badge)*
 - **40% ≤ Risk < 65%:** *SUSPICIOUS (Step-up biometric verification, yellow alert)*
 - **Risk ≥ 65%:** *CRITICAL CLONING ATTACK (Immediate hangup, SIP drop, SMS/Email dispatch)*